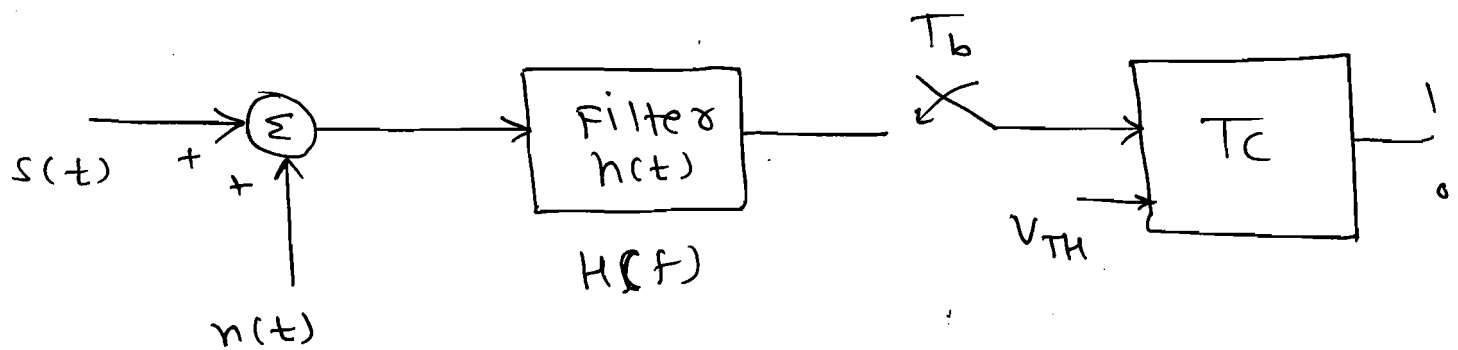


★ Matched Filter Receiver :-

⇒



⇒ The received signal and noise is applied to the filter whose impulse response is $h(t)$. The output of the filter is sampled at every T_b sec. The sampled value is compared with the Threshold voltage to take decision whether the received signal is '0' (or) '1'. If effect of the noise is very large then binary symbol '1' will be received as '0' and '0' will be received as '1'. This error should be as minimum as possible. So, the SNR should be as high as possible.

⇒ The signal at the o/p of the

$$S_0(t) = s(t) \otimes h(t).$$

$$\Rightarrow S_0(t) = s(t) \cdot h(t).$$

$$\therefore S_0(t) = \int_{-\infty}^{\infty} s(\tau) \cdot h(t) \cdot e^{j2\pi\tau t} \cdot d\tau.$$

$\Rightarrow$ Assume that the old signal is sample

at $t = T_b$ sec.

$$\text{So, } S_0(T_b) = \int_{-\infty}^{\infty} s(\tau) \cdot h(\tau) \cdot e^{j2\pi\tau T_b} \cdot d\tau.$$

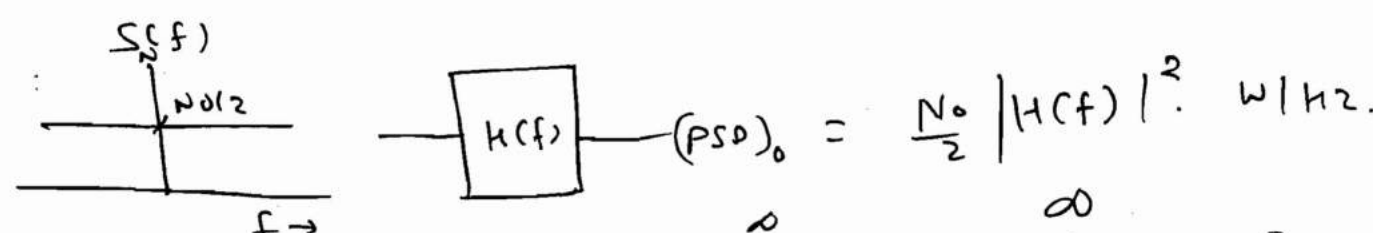
$\Rightarrow$ The signal power at the threshold

computer is,

$$\text{signal power} = |S_0(T_b)|^2$$

$$= \left| \int_{-\infty}^{\infty} s(\tau) \cdot h(\tau) \cdot e^{j2\pi\tau T_b} \cdot d\tau \right|^2$$

$\Rightarrow$



$$(PSD)_0 = \frac{N_0}{2} |H(f)|^2 \text{ W/Hz.}$$

$$\text{Power} = \int_{-\infty}^{\infty} (PSD)_0 \cdot df = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \cdot df.$$

$$\Rightarrow (S/N) = \left| \int_{-\infty}^{\infty} H(f) \cdot s(\tau) \cdot e^{j2\pi\tau T_b} \cdot df \right|^2$$

* Schwarz inequality:

$$\left| \int_{-\infty}^{\infty} f_1(t) \cdot f_2(t) \cdot dt \right|^2 \leq \int_{-\infty}^{\infty} |f_1(t)|^2 \cdot dt \cdot \int_{-\infty}^{\infty} |f_2(t)|^2 \cdot dt$$

$$\text{LHS} = \text{RHS}$$

only when $f_1(t) = f_2^*(t)$.

$$\therefore (S/N) \leq \frac{\int_{-\infty}^{\infty} |H(f)|^2 \cdot df \cdot \int_{-\infty}^{\infty} |S(f) \cdot e^{j2\pi f T_b}|^2 \cdot df}{\int_{-\infty}^{\infty} |H(f)|^2 \cdot df}$$

$$\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \cdot df$$

$$\therefore (S/N) \leq \frac{\int_{-\infty}^{\infty} |S(f)|^2 \cdot df}{\frac{N_0}{2}}$$

$$(\because |e^{j2\pi f T_b}| = 1)$$

$$S(t) \rightarrow S(f)$$

$$|S(f)|^2 \rightarrow \text{ESD}$$

$$\int_{-\infty}^{\infty} |S(f)|^2 \cdot df \rightarrow E$$

$$\therefore (S/N) \leq \frac{E}{(N_0/2)}$$

$$\therefore \boxed{(S/N)_{\max} = \frac{2E}{N_0}} \quad \text{IMP}$$

only when $H(f) = S^*(f) \cdot e^{-j2\pi f T_b}$

$$\Rightarrow h(t) = \int_{-\infty}^{\infty} H(f) \cdot e^{j2\pi ft} \cdot df.$$

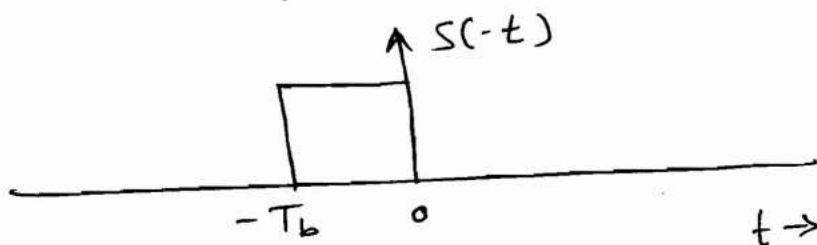
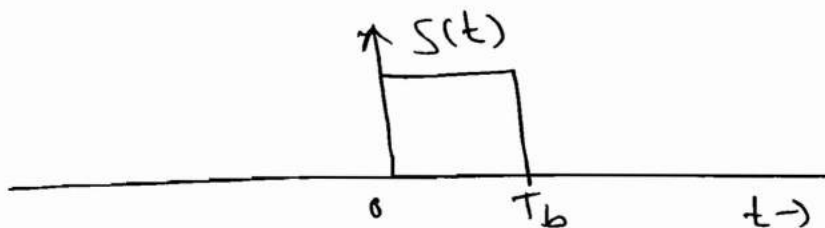
$$\therefore h(t) = \int_{-\infty}^{\infty} S^*(f) \cdot e^{-j2\pi f T_b} \cdot e^{+j2\pi f t} \cdot df$$

$$\therefore h(t) = \int_{-\infty}^{\infty} S^*(f) \cdot e^{-j2\pi f (T_b - t)} \cdot df.$$

$$h(t) = \left[\int_{-\infty}^{\infty} S(f) \cdot e^{j2\pi f (T_b - t)} \cdot df \right]^*$$

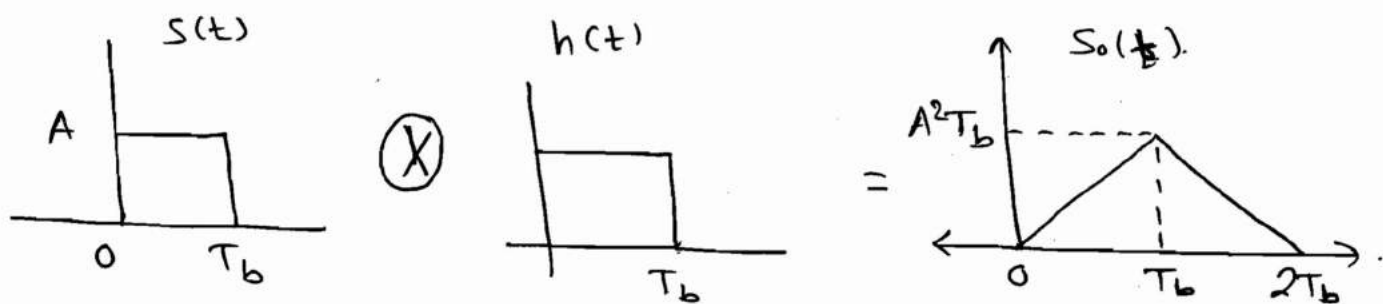
$$\therefore h(t) = S^*(T_b - t).$$

$\Rightarrow$ To determine the impulse response of the filter the following procedure is used:



$\Rightarrow$ The impulse response ^{of the filter} is same as the impulse input signal (or) impulse response of the filter is matched to ip signal. ~~therefore~~ That's why it is called as Matched filter receiver.

$\Rightarrow$ The output of the filter is,



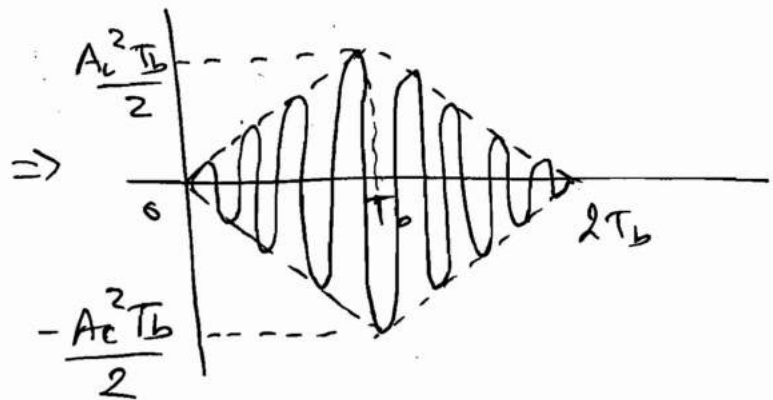
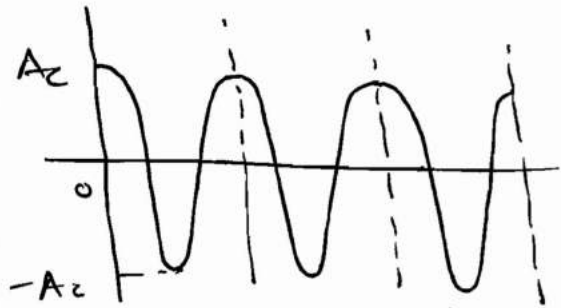
$\Rightarrow$ The Sampled value (or) the max. value is always equal to the energy of signal,

$$E = \int_0^{T_b} A^2 dt.$$

$\Rightarrow$ The input to the Threshold Comparator is either $A^2 T_b$ (or) 0 when ON - OFF signaling is used.

$\Rightarrow$ The input to the Threshold Comparator is $A^2 T_b$ (or) $-A^2 T_b$ when NRZ signaling

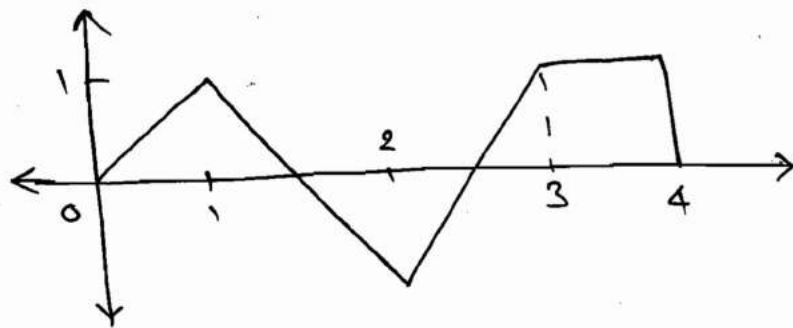
$\Rightarrow$ Assume that the i/p to the matched filter is $A_c \cos 2\pi f_c t$ where,
 $T_b = N/f_c$ where N is integer.



$\Rightarrow$ The impulse response of the filter will be matched to the input signal only when Rectangular pulse and carrier.

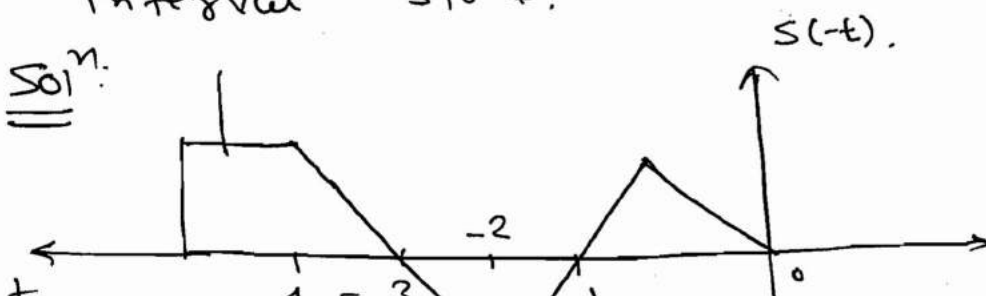
Q The i/p to the matched filter is as shown in the figure.

Gate-2004

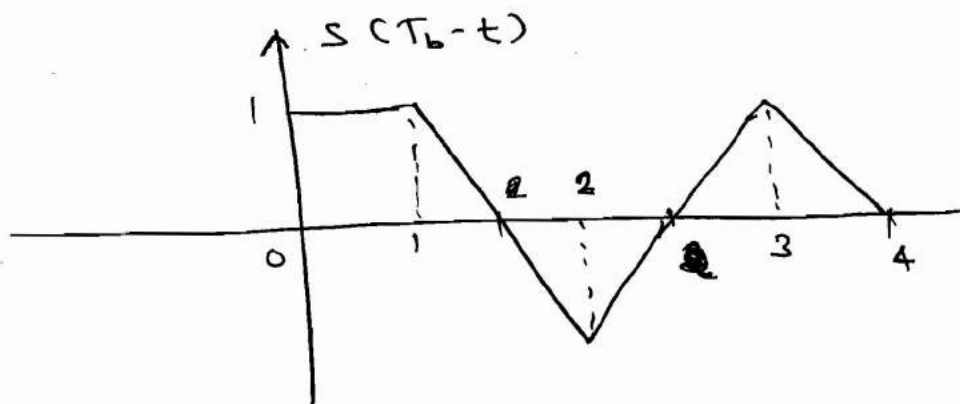


determine the slope of $h(t)$ in the interval 3 to 4.

Solⁿ:



$\Rightarrow$

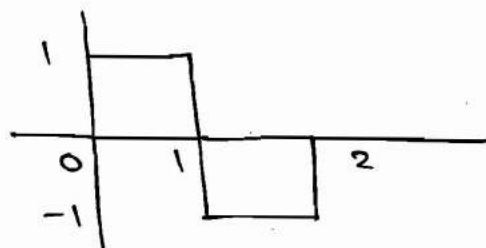


$$\text{Slope} = \frac{1-0}{2-1} = -1$$

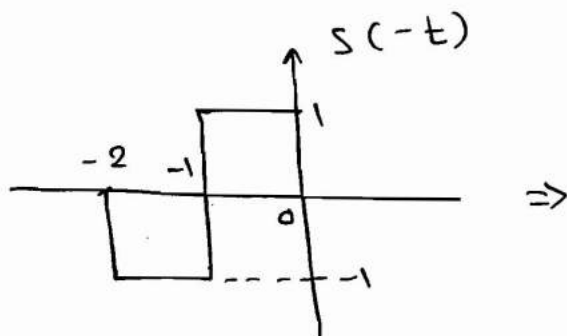
Q-2

The i/p to the matched filter is shown in figure. determine the h(t).

s



Soln:



$\Downarrow$

